# COMSATS University Islamabad Wah Campus




## Department of Computer Science

### Internship Report (Week No. 1)

## Submitted By :

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**Submission Date : 10 August 2026 (Monday)**

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# CHAPTER 1 : INTRODUCTION TO DATABASE ADMINISTRATION (DBA)

## 1.1 Introduction to Database Administration (DBA)

Database Administration (DBA) is the process of managing, maintaining, monitoring, and securing databases to ensure that data remains accurate, available, and protected. A Database Administrator (DBA) is responsible for installing database software, configuring database environments, managing users, performing backups and recovery, monitoring database performance and ensuring database security.

Nowadays, every organization relies on databases to store and manage its valuable information. Therefore, Database Administration has become one of the most important areas of Information Technology.

## 1.2 History of Database Administration

Initially, organizations maintained their records manually in paper files. As technology evolved, File Management Systems were introduced, but these systems had several limitations such as data redundancy, inconsistency and poor security.

To overcome these issues, Database Management Systems (DBMS) were developed. Later, relational databases became the industry standard, and organizations started using enterprise database systems such as Oracle Database. As databases became larger and more complex, the role of the Database Administrator (DBA) was introduced to manage and maintain these systems efficiently.

## 1.3 Need of Database Administration

Database Administration is required to ensure that databases remain secure, efficient and continuously available. The major objectives of DBA include:

- Managing organizational data efficiently.
- Maintaining data security.
- Performing backup and recovery.
- Monitoring database performance.
- Managing database users and privileges.
- Preventing unauthorized access.
- Ensuring data consistency and integrity.
- Supporting continuous business operations.

## 1.4 Roles and Responsibilities of a DBA

The responsibilities of a Database Administrator include:

- Installing and configuring database software.
- Creating and managing databases.
- Managing database users and permissions.
- Performing backup and recovery operations.

- Monitoring database performance.
- Managing storage and tablespaces.
- Applying updates and patches.
- Troubleshooting database-related issues.
- Ensuring database security and availability.
- Maintaining technical documentation.

### 1.5 Market Scope of DBA

**Database Administration** is one of the most demanding career fields in Information Technology. Every organization that stores digital data requires skilled database professionals. Database Administrators are employed in:

- Government Organizations
- Banking Sector
- Software Houses
- Hospitals
- Educational Institutions
- Telecommunication Companies
- Manufacturing Industries
- E-Commerce Companies
- Cloud Service Providers

Oracle Database Administrators are especially in demand because Oracle Database is widely used in enterprise-level organizations. With experience and certifications, a DBA can advance to senior technical and management positions.

# CHAPTER 2 : VIRTUALIZATION AND ORACLE LINUX ENVIRONMENT SETUP

## 2.1 Practical No. 1: Installation of VMware Workstation

### 2.1.1 Objective

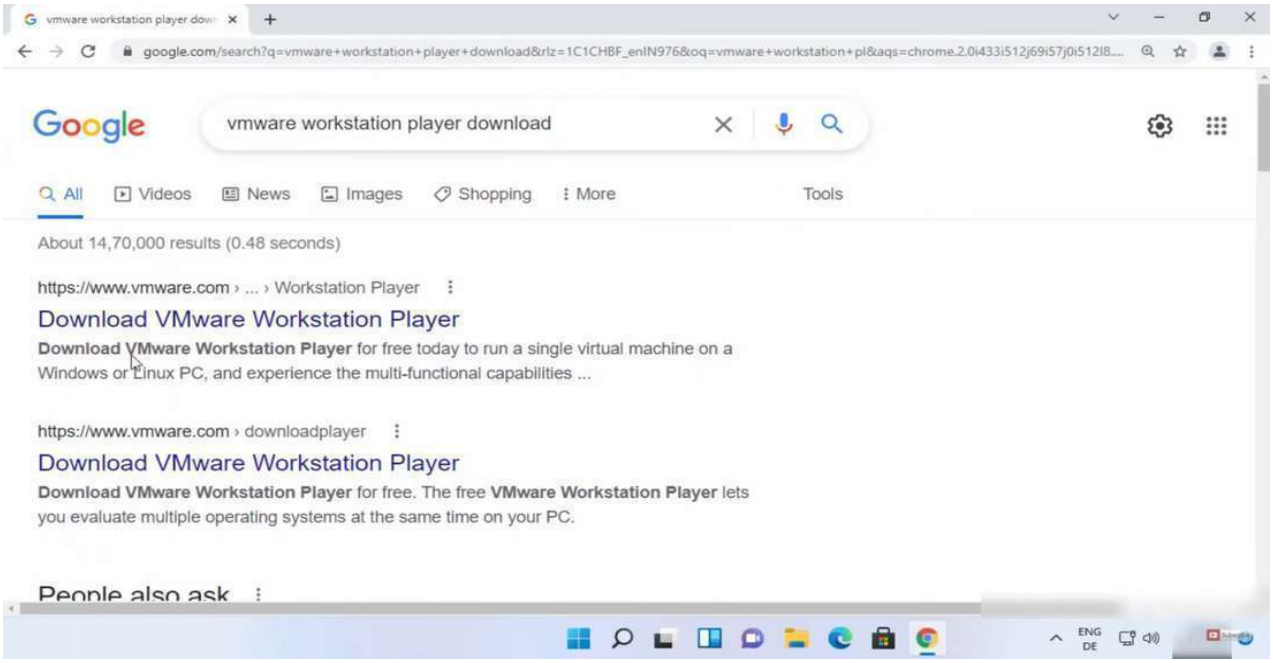
The objective of this practical was to install VMware Workstation for creating a virtual environment where Oracle Linux and Oracle Database could be installed and managed without affecting the host operating system.

### 2.1.2 Step 1: Download VMware Workstation

The first step was to download the VMware Workstation installer from the official VMware website. VMware Workstation is a virtualization software that allows multiple operating systems to

run on a single physical computer. This software was required to create a virtual machine for Oracle Linux.

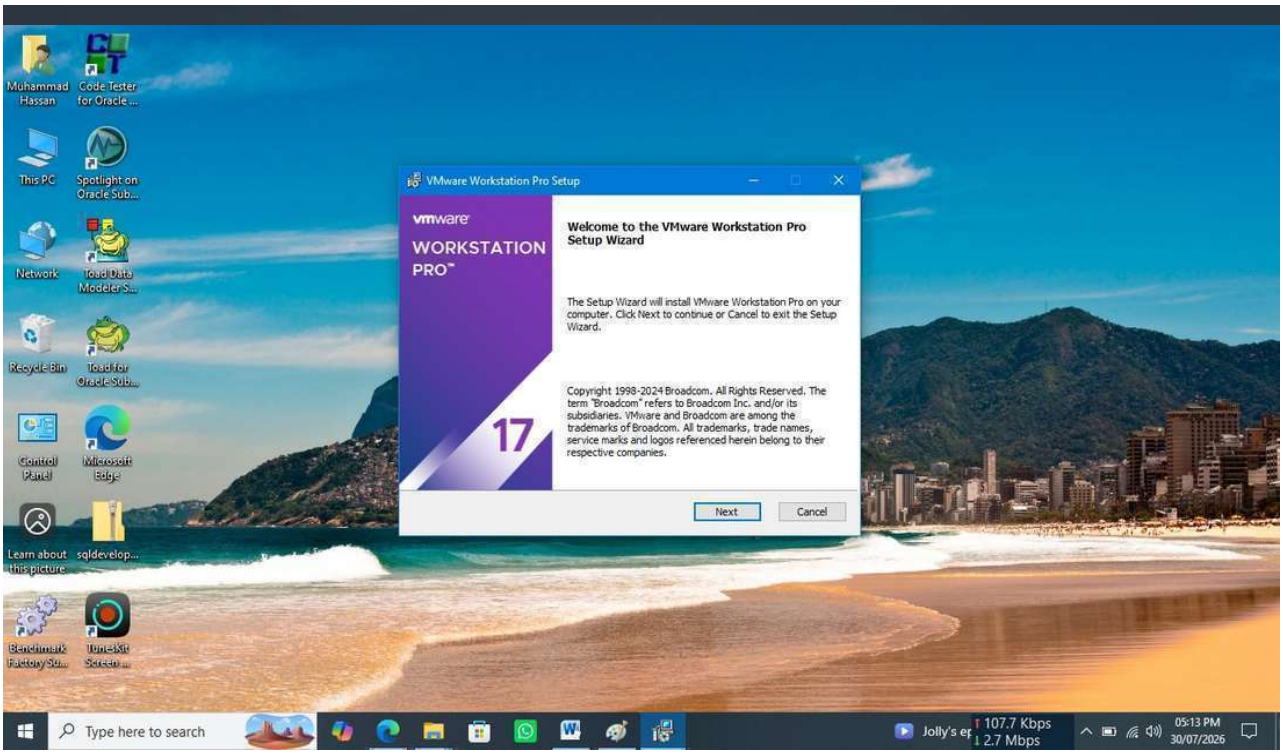
**Figure 2.1: Downloaded VMware Workstation Installer**




### 2.1.3 Step 2: Run the VMware Installer

After downloading the installer, it was executed with administrator privileges. The installation wizard guided the installation process by allowing the required options to be selected. The default settings were used as they are suitable for most users.

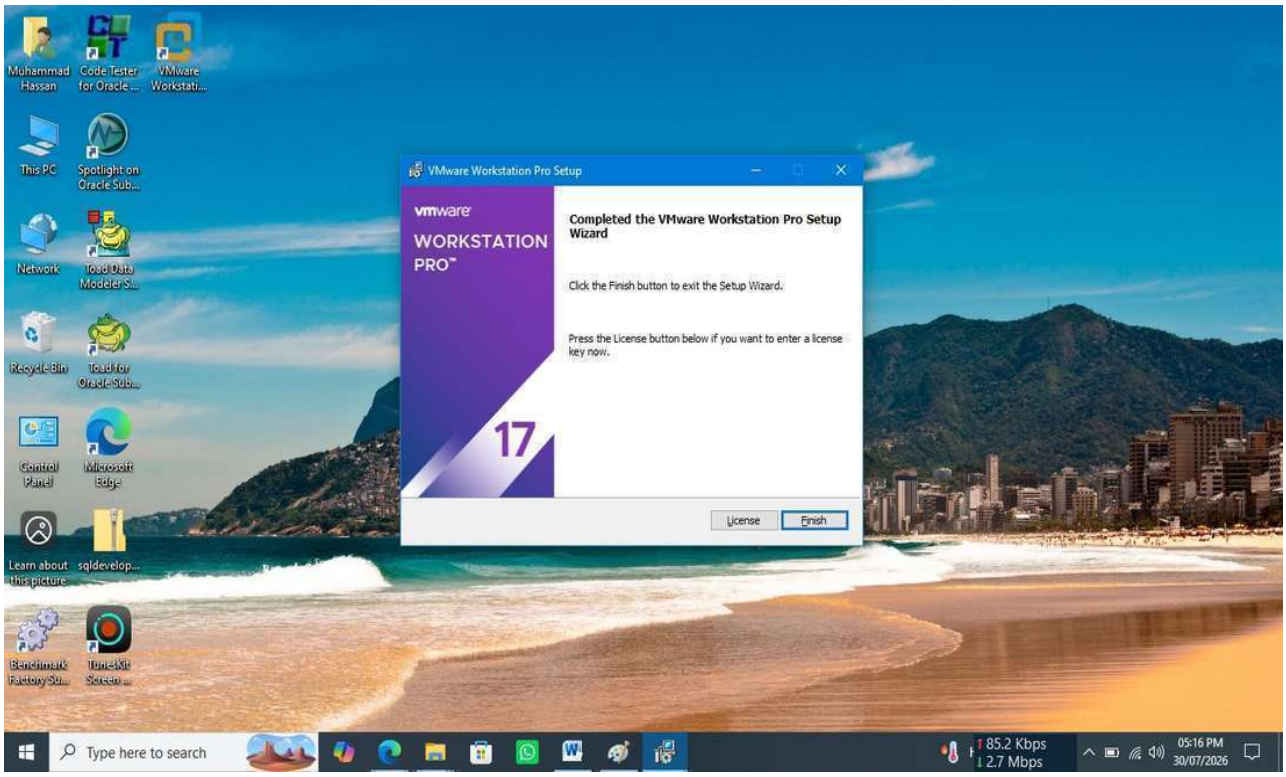
**Figure 2.2: VMware Installation Wizard**




### 2.1.4 Step 3: Complete the Installation

The installation process copied all the necessary files to the system and configured VMware Workstation. After the installation was completed successfully, the software was ready for use.

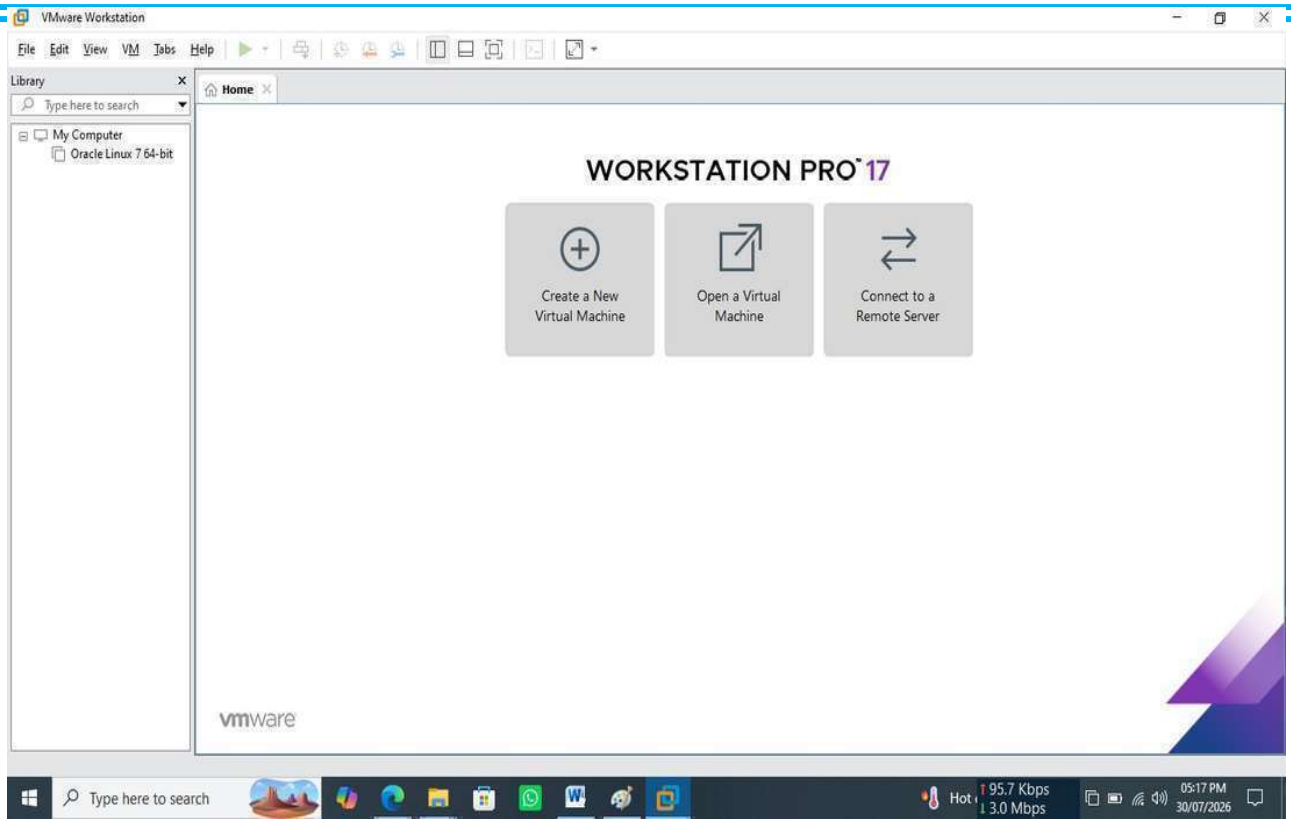
**Figure 2.3: VMware Workstation Successfully Installed**




### 2.1.5 Step 4: Verify the Installation

VMware Workstation was launched to verify that the installation had completed successfully. The successful appearance of the VMware home screen confirmed that the software was working properly.

**Figure 2.4: VMware Workstation Home Screen**




## 2.2 Practical No. 2: Installation of Oracle Linux on VMware

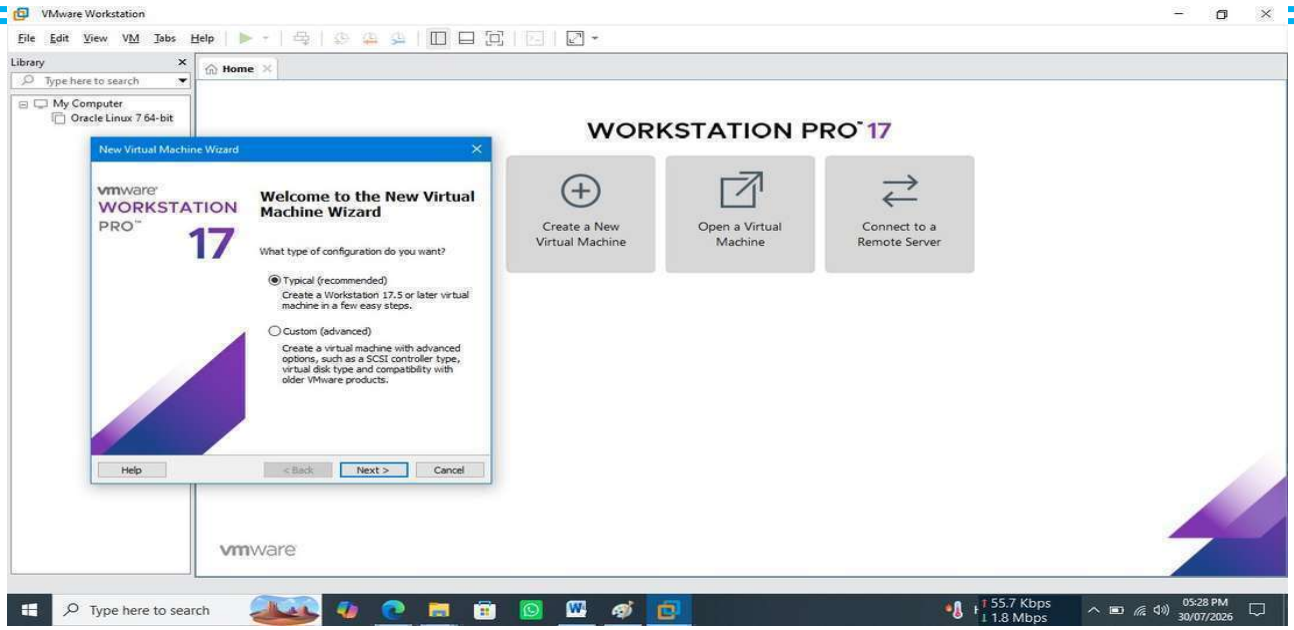
### 2.2.1 Objective

The objective of this practical was to install Oracle Linux inside VMware Workstation. Oracle Linux serves as the operating system required for installing and managing Oracle Database during the internship.

### 2.2.2 Step 1: Create a New Virtual Machine

A new virtual machine was created in VMware Workstation. During this process, the virtual hardware configuration was defined, allowing Oracle Linux to run independently from the host operating system.

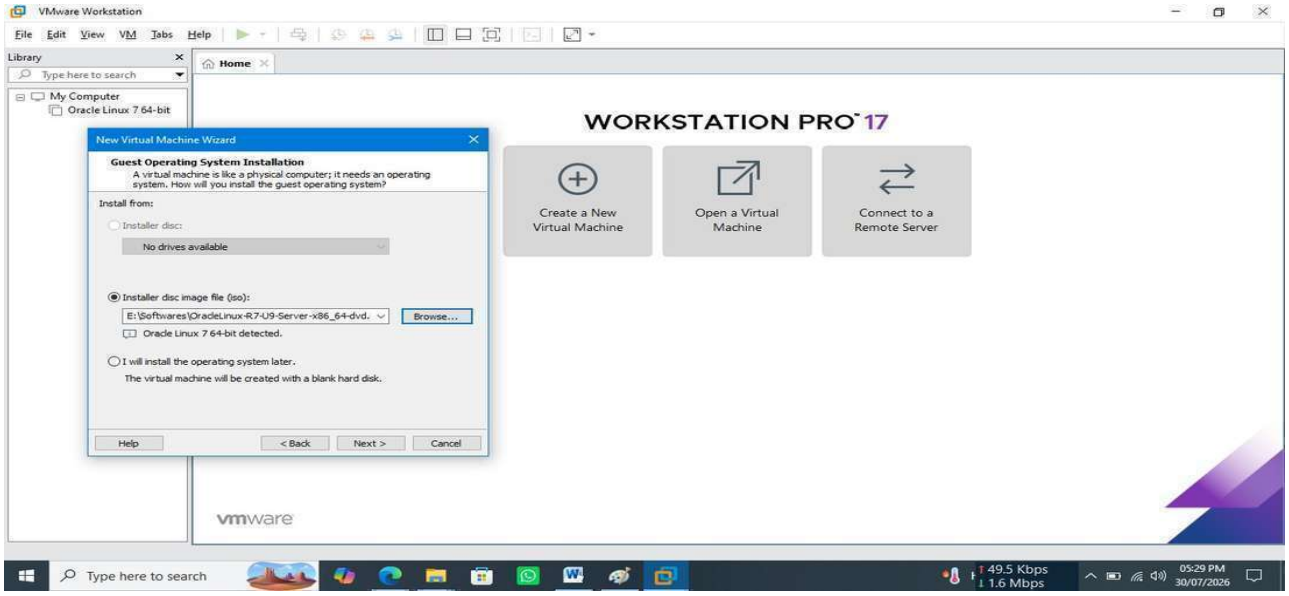
**Figure 2.5: Creating a New Virtual Machine**




### 2.2.3 Step 2: Select the Oracle Linux ISO File

The Oracle Linux ISO image was selected as the installation media. This ISO file contains the complete operating system files required for installation inside the virtual machine.

**Figure 2.6: Oracle Linux ISO Selection**

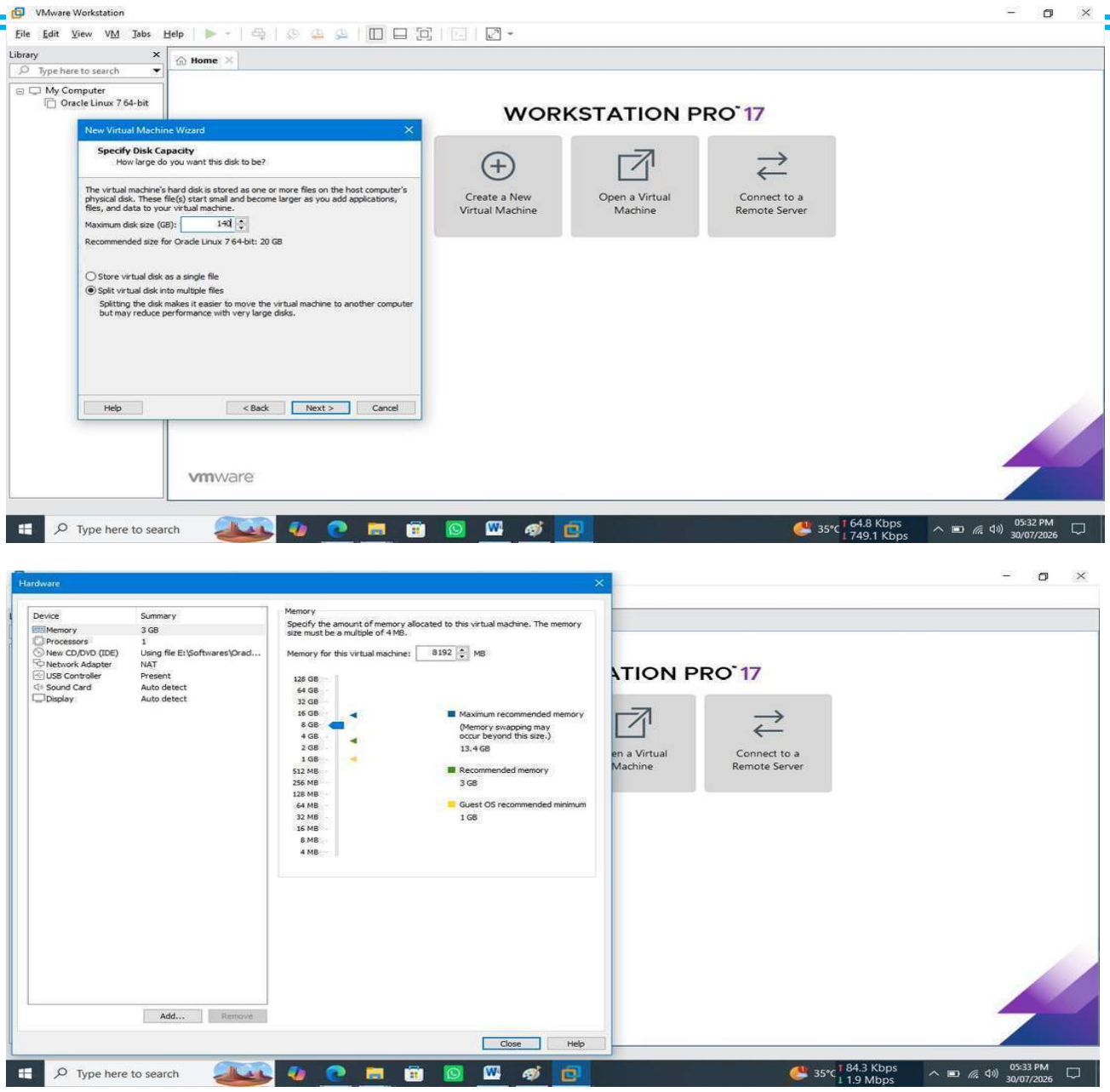



### 2.2.4 Step 3: Configure Virtual Machine Hardware

The virtual machine was configured by allocating RAM, processor cores, and hard disk space according to the system requirements. Proper hardware allocation is important to ensure smooth performance during Oracle Database installation.

- RAM: __8__ GB
- Processor: ___8___ Cores
- Hard Disk: ___150___ GB

**Figure 2.7: Virtual Machine Hardware Configuration**



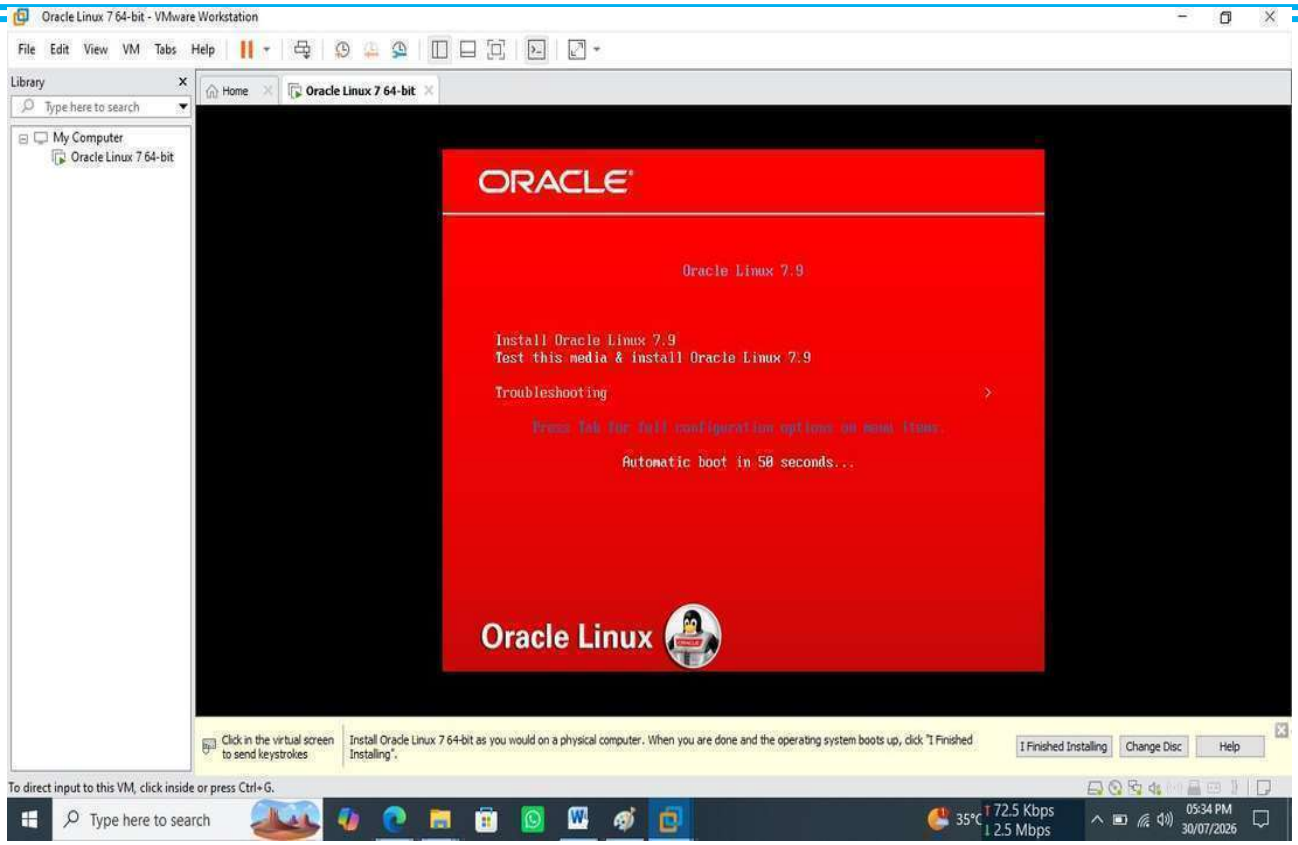




### 2.2.5 Step 4: Install Oracle Linux

The Oracle Linux installation process was started by selecting the installation language and configuring the installation destination. The hostname, root password and user account were also configured before beginning the installation.

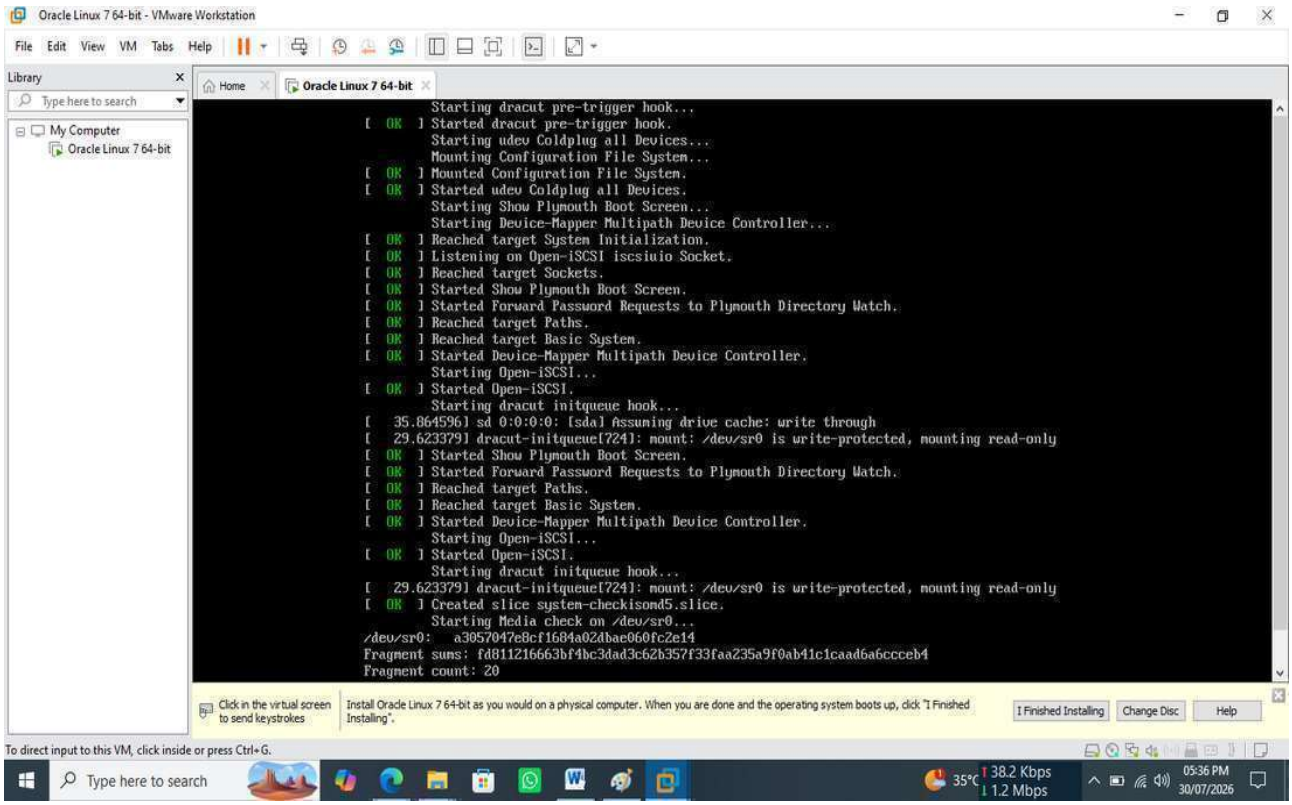
**Figure 2.8: Oracle Linux Installation Screen**

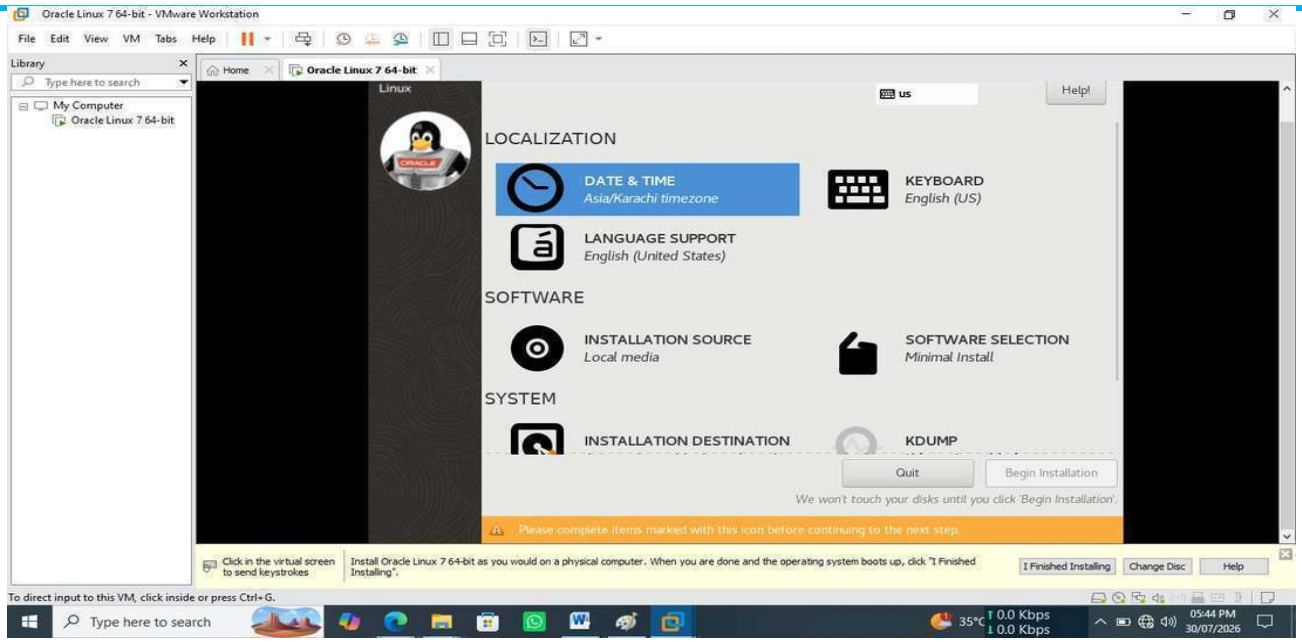


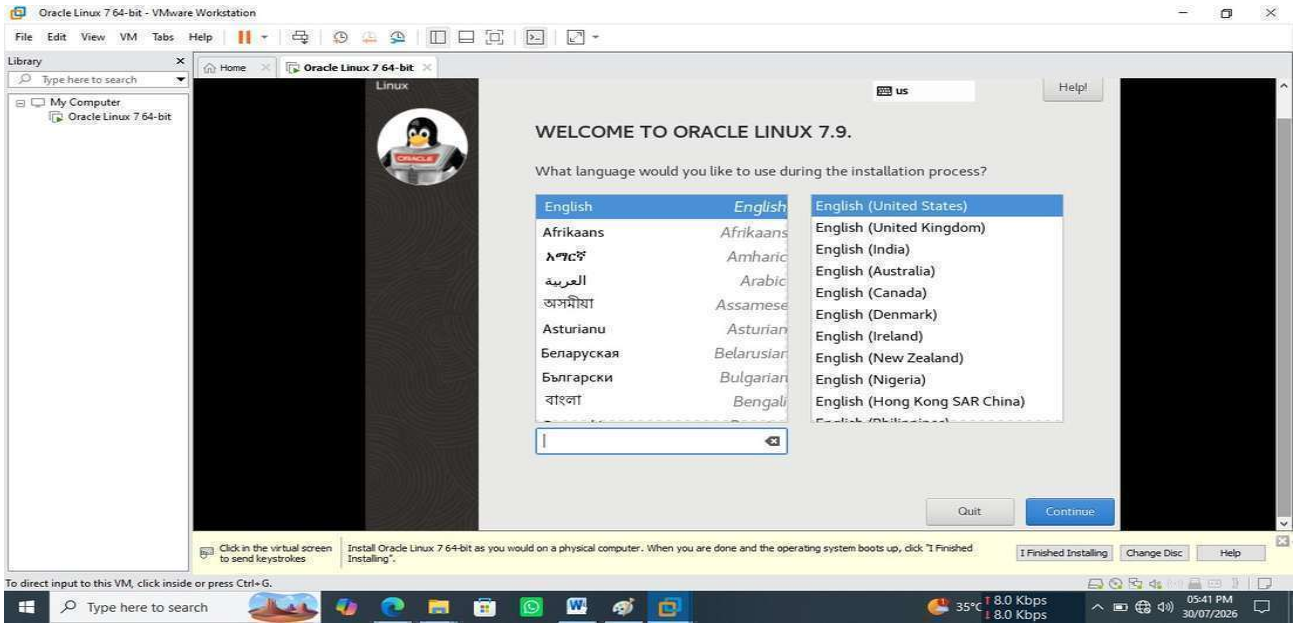

### 2.2.6 Step 5: Installation Progress

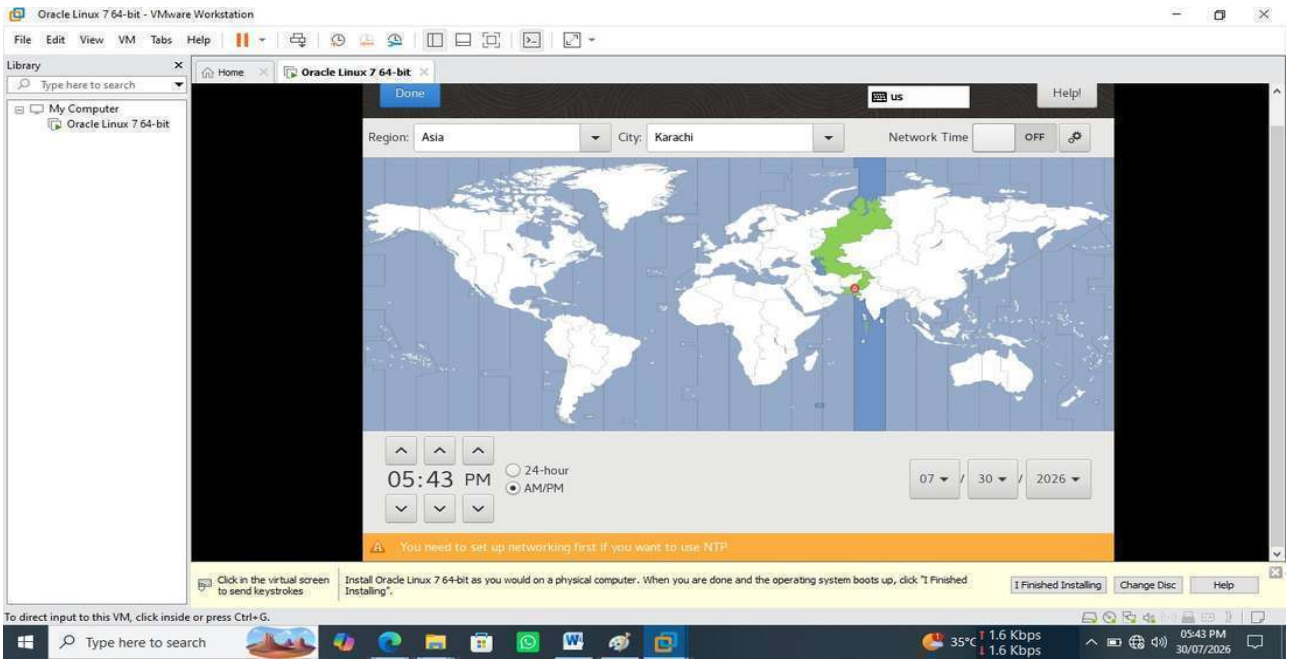
During this stage, Oracle Linux copied the required system files and installed all necessary components. The installation progress was monitored until it reached completion.

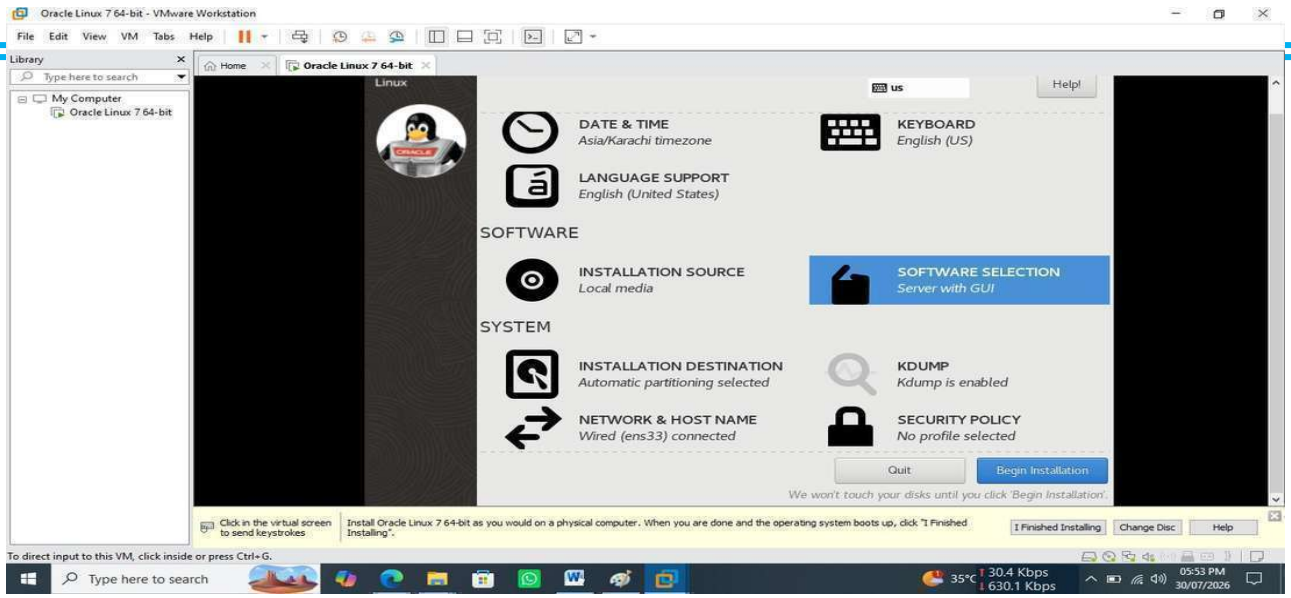
**Figure 2.9: Oracle Linux Installation Progress**




Oracle Linux 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Oracle Linux 7 64-bit
Home
Oracle Linux 7 64-bit
Linux
us
Help!
LOCALIZATION
DATE & TIME
Asia/Karachi timezone
KEYBOARD
English (US)
LANGUAGE SUPPORT
English (United States)
SOFTWARE
INSTALLATION SOURCE
Local media
SOFTWARE SELECTION
Minimal Install
SYSTEM
INSTALLATION DESTINATION
KDUMP
Quit
Begin Installation
We won't touch your disks until you click 'Begin Installation'.
Please complete items marked with this icon before continuing to the next step.
Click in the virtual screen to send keystrokes
Install Oracle Linux 7 64-bit as you would on a physical computer. When you are done and the operating system boots up, click "I Finished Installing".
I Finished Installing
Change Disc
Help
To direct input to this VM, click inside or press Ctrl+G.
Type here to search
35°C
0.0 Kbps
0.0 Kbps
05:44 PM
30/07/2026

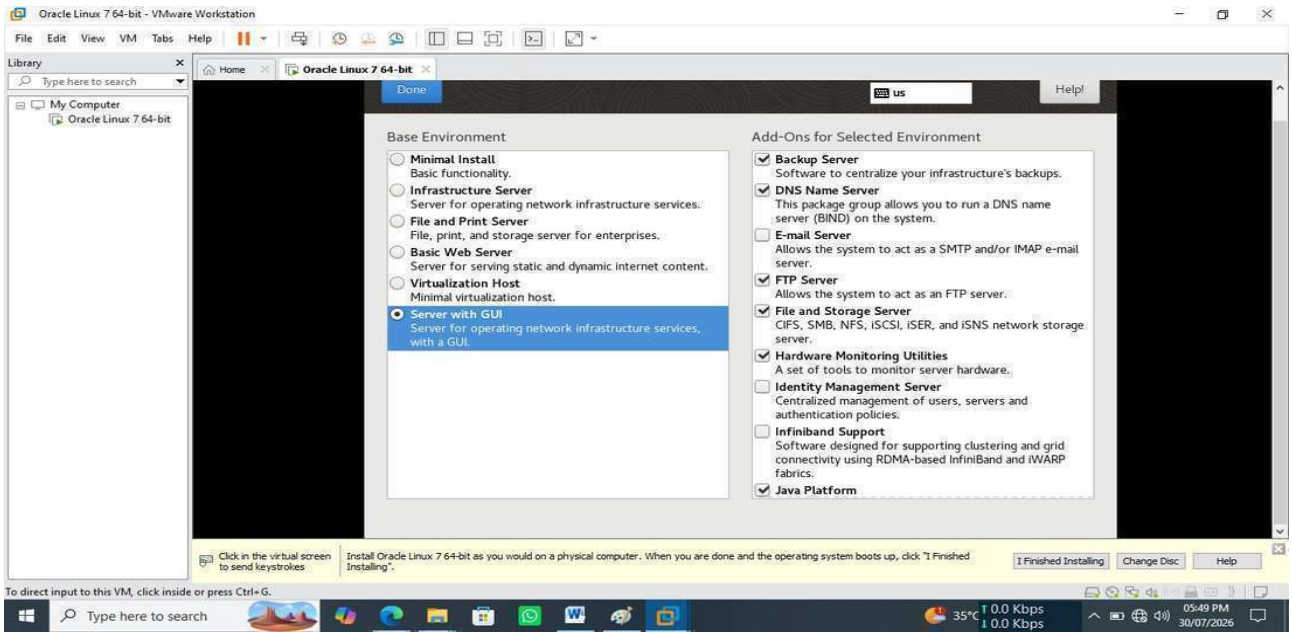

Oracle Linux 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Oracle Linux 7 64-bit
Home
Oracle Linux 7 64-bit
Linux
us
Help!
WELCOME TO ORACLE LINUX 7.9.
What language would you like to use during the installation process?
English
English
Afrikaans
Afrikaans
አማርኛ
Amharic
العربية
Arabic
অসমীয়া
Assamese
Asturianu
Asturian
Беларуская
Belarusian
Български
Bulgarian
বাংলা
Bengali
English (United States)
English (United Kingdom)
English (India)
English (Australia)
English (Canada)
English (Denmark)
English (Ireland)
English (New Zealand)
English (Nigeria)
English (Hong Kong SAR China)
Quit
Continue
Click in the virtual screen to send keystrokes
Install Oracle Linux 7 64-bit as you would on a physical computer. When you are done and the operating system boots up, click "I Finished Installing".
I Finished Installing
Change Disc
Help
To direct input to this VM, click inside or press Ctrl+G.
Type here to search
35°C
8.0 Kbps
8.0 Kbps
05:41 PM
30/07/2026

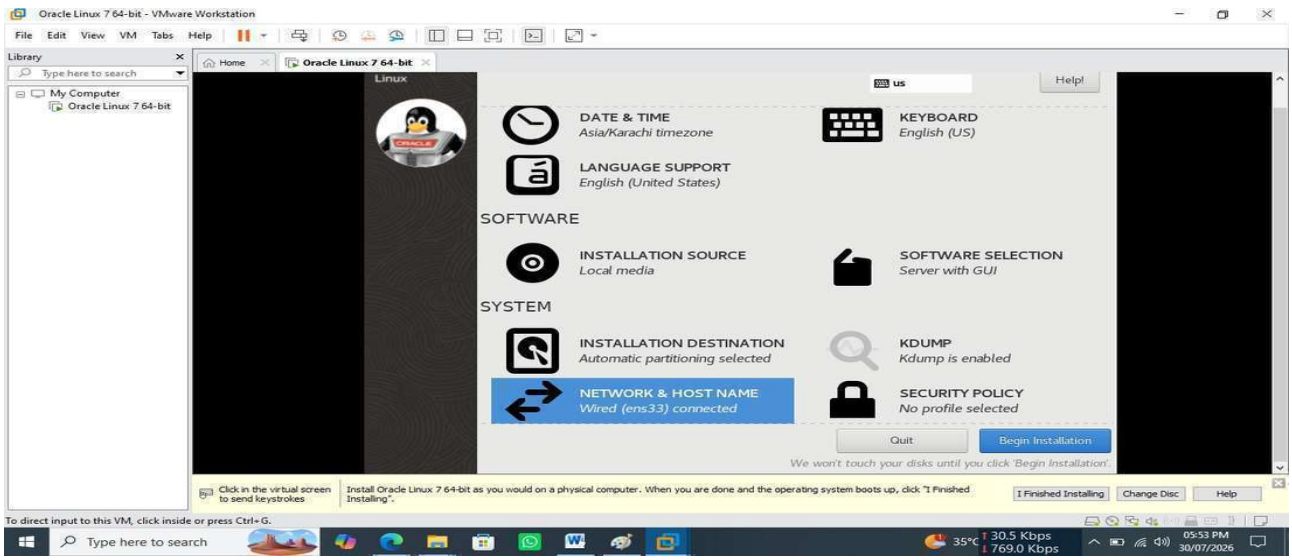

Oracle Linux 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Oracle Linux 7 64-bit
Home
Oracle Linux 7 64-bit
Done
us
Help!
Region: Asia
City: Karachi
Network Time
OFF
05:43 PM
24-hour
AM/PM
07
30
2026
You need to set up networking first if you want to use NTP
Click in the virtual screen to send keystrokes
Install Oracle Linux 7 64-bit as you would on a physical computer. When you are done and the operating system boots up, click "I Finished Installing".
I Finished Installing
Change Disc
Help
To direct input to this VM, click inside or press Ctrl+G.
Type here to search
35°C
1.6 Kbps
1.6 Kbps
05:43 PM
30/07/2026

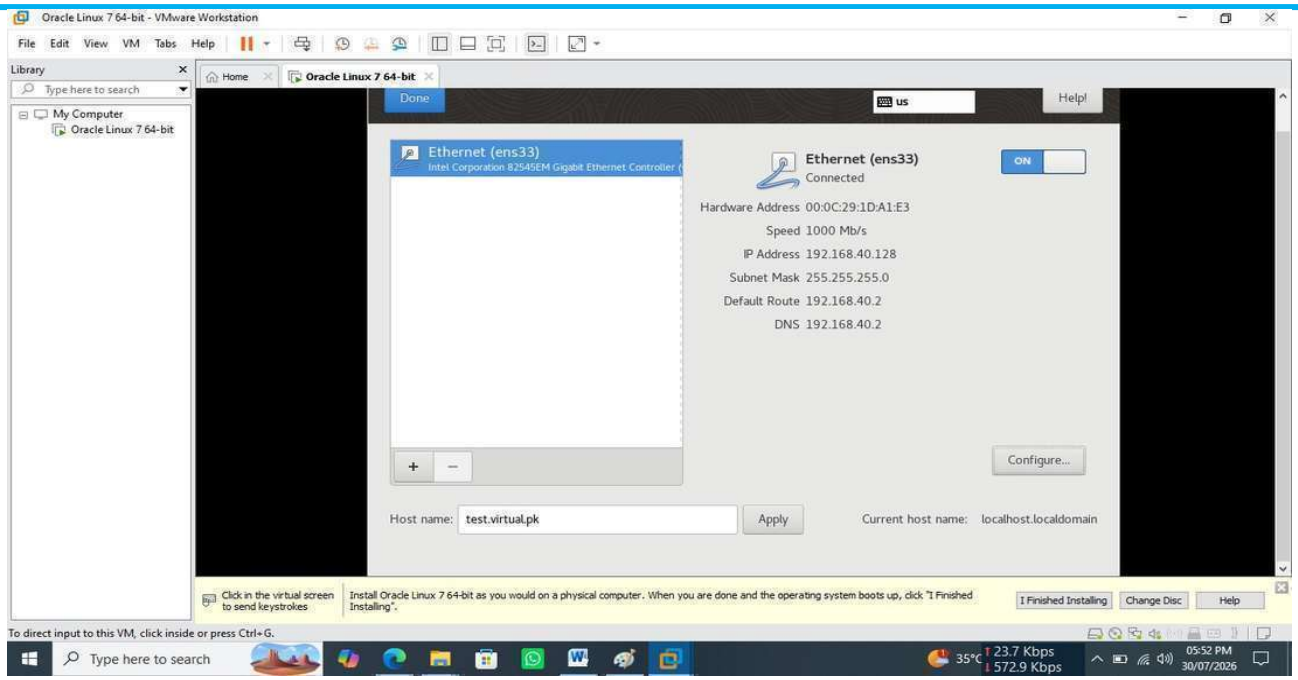


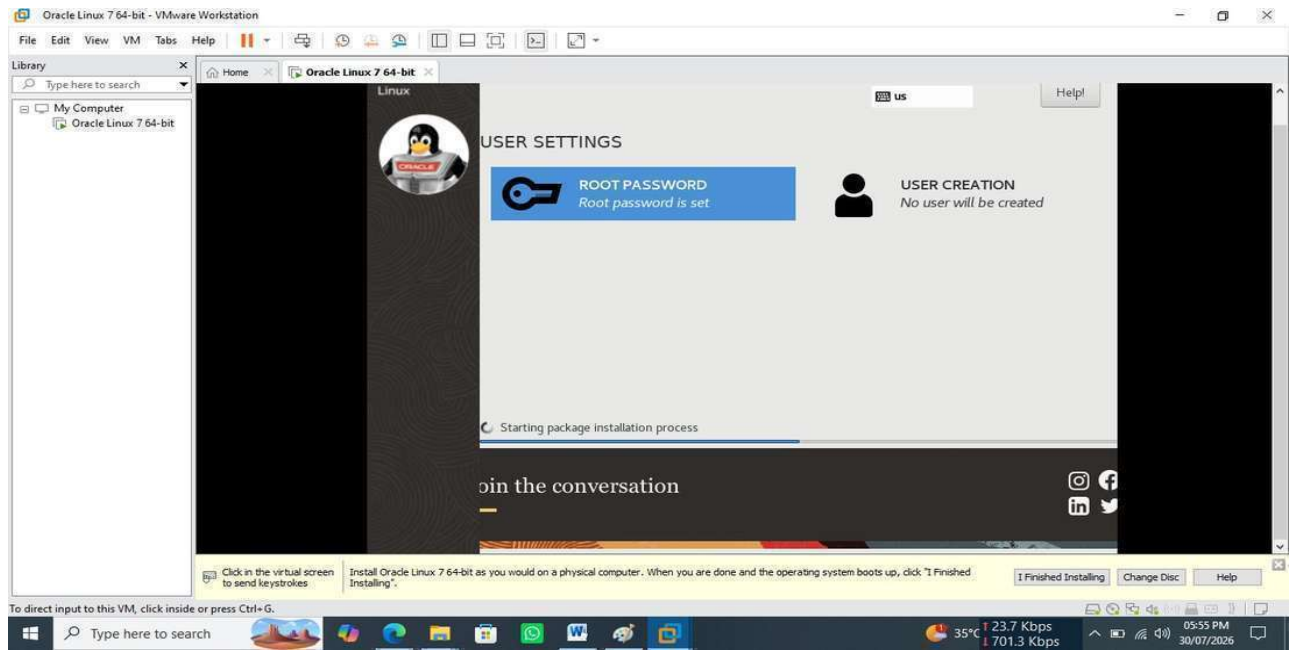

- **Add-Ons For Selected Environments : (Do Not Marked Following Options)**

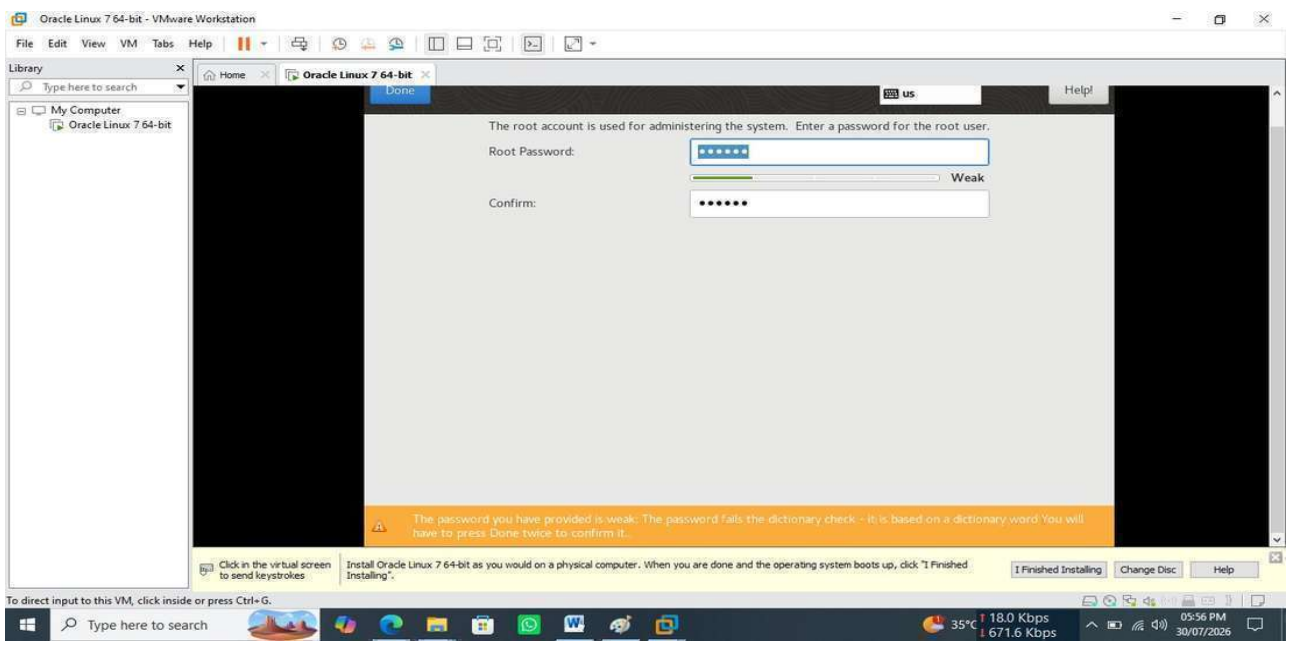
Email Server, Identity Management Server, Infiniband Support, Print Server, PostgreSQL Database Server, MariaDB Database Server, Load Balancer, Smart Card Support.

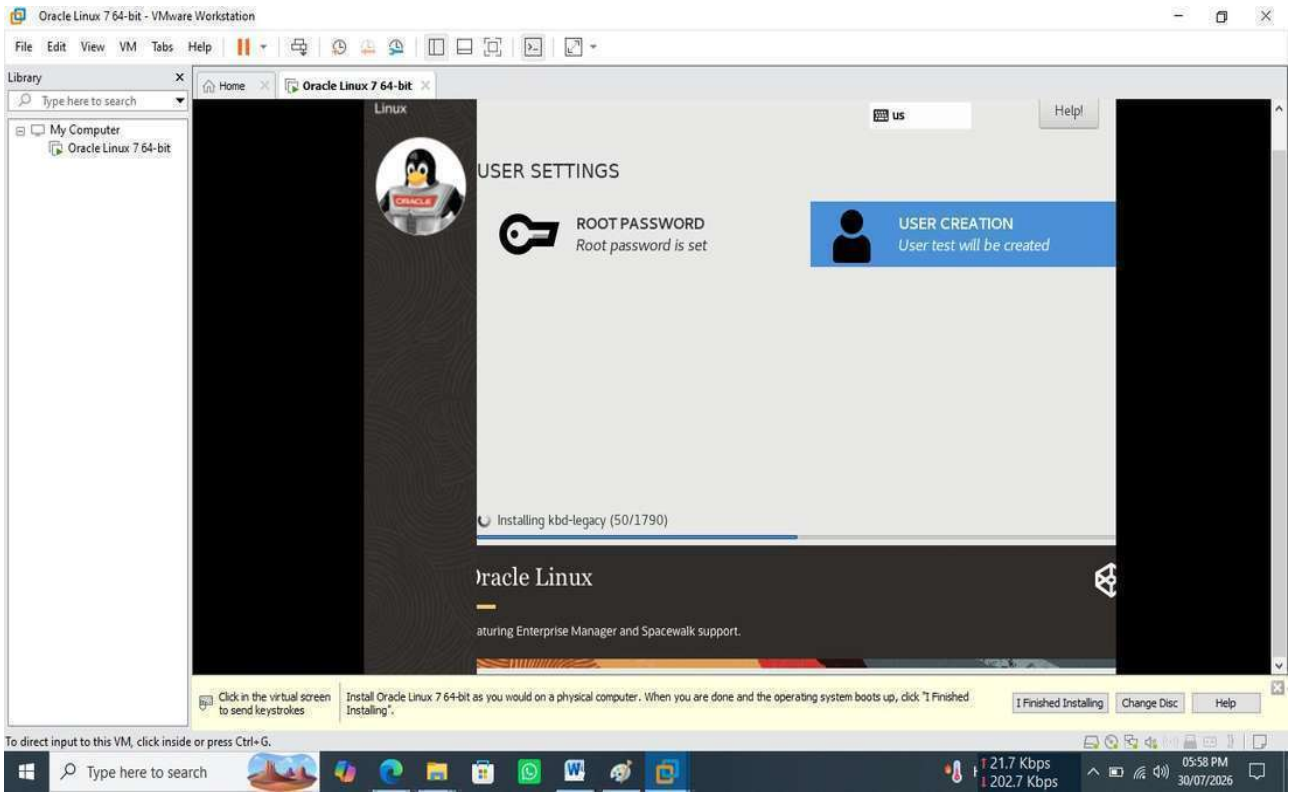


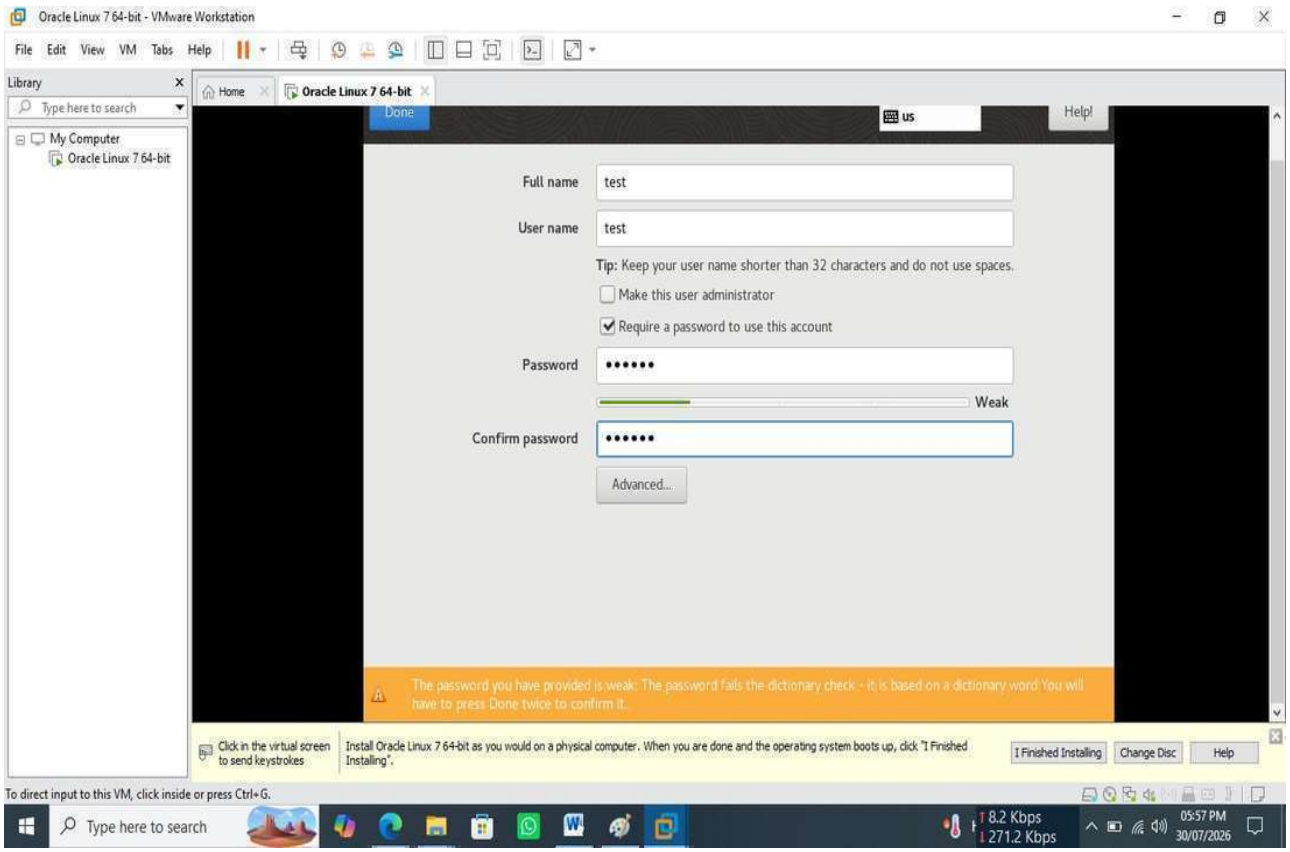




Oracle Linux 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Oracle Linux 7 64-bit
Home
Oracle Linux 7 64-bit
Done
us
Help!
Ethernet (ens33)
Intel Corporation 82545EM Gigabit Ethernet Controller
Ethernet (ens33)
Connected
ON
Hardware Address 00:0C:29:1D:A1:E3
Speed 1000 Mb/s
IP Address 192.168.40.128
Subnet Mask 255.255.255.0
Default Route 192.168.40.2
DNS 192.168.40.2
Configure...
Host name: test.virtual.pk
Apply
Current host name: localhost.localdomain
Click in the virtual screen to send keystrokes
Install Oracle Linux 7 64-bit as you would on a physical computer. When you are done and the operating system boots up, click "I Finished Installing".
I Finished Installing
Change Disc
Help
To direct input to this VM, click inside or press Ctrl+G.
Type here to search
35°C
23.7 Kbps
572.9 Kbps
05:52 PM
30/07/2026


Oracle Linux 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Oracle Linux 7 64-bit
Home
Oracle Linux 7 64-bit
Linux
us
Help!
USER SETTINGS
ROOT PASSWORD
Root password is set
USER CREATION
No user will be created
Starting package installation process
oin the conversation
Click in the virtual screen to send keystrokes
Install Oracle Linux 7 64-bit as you would on a physical computer. When you are done and the operating system boots up, click "I Finished Installing".
I Finished Installing
Change Disc
Help
To direct input to this VM, click inside or press Ctrl+G.
Type here to search
35°C
23.7 Kbps
701.3 Kbps
05:55 PM
30/07/2026


Oracle Linux 7 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Oracle Linux 7 64-bit
Home
Oracle Linux 7 64-bit
Done
us
Help!
The root account is used for administering the system. Enter a password for the root user.
Root Password:
Weak
Confirm:
The password you have provided is weak: The password fails the dictionary check - it is based on a dictionary word. You will have to press Done twice to confirm it.
Click in the virtual screen to send keystrokes
Install Oracle Linux 7 64-bit as you would on a physical computer. When you are done and the operating system boots up, click "I Finished Installing".
I Finished Installing
Change Disc
Help
To direct input to this VM, click inside or press Ctrl+G.
Type here to search
35°C
18.0 Kbps
671.6 Kbps
05:56 PM
30/07/2026




### 2.2.7 Step 6: Restart and Log In

After the installation was completed, the virtual machine was restarted. The newly created user account was used to log in to Oracle Linux, confirming that the operating system had been installed successfully.

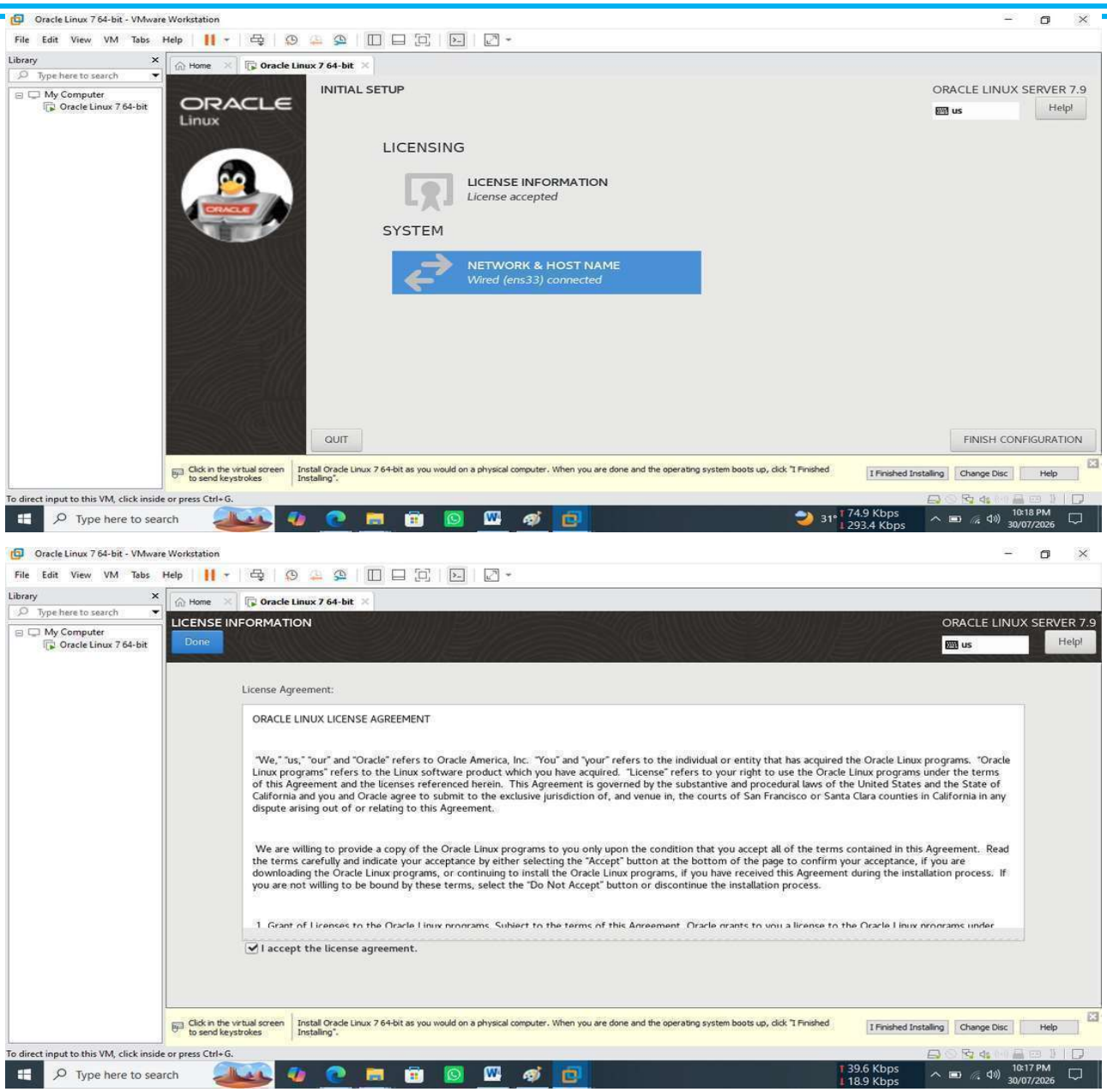
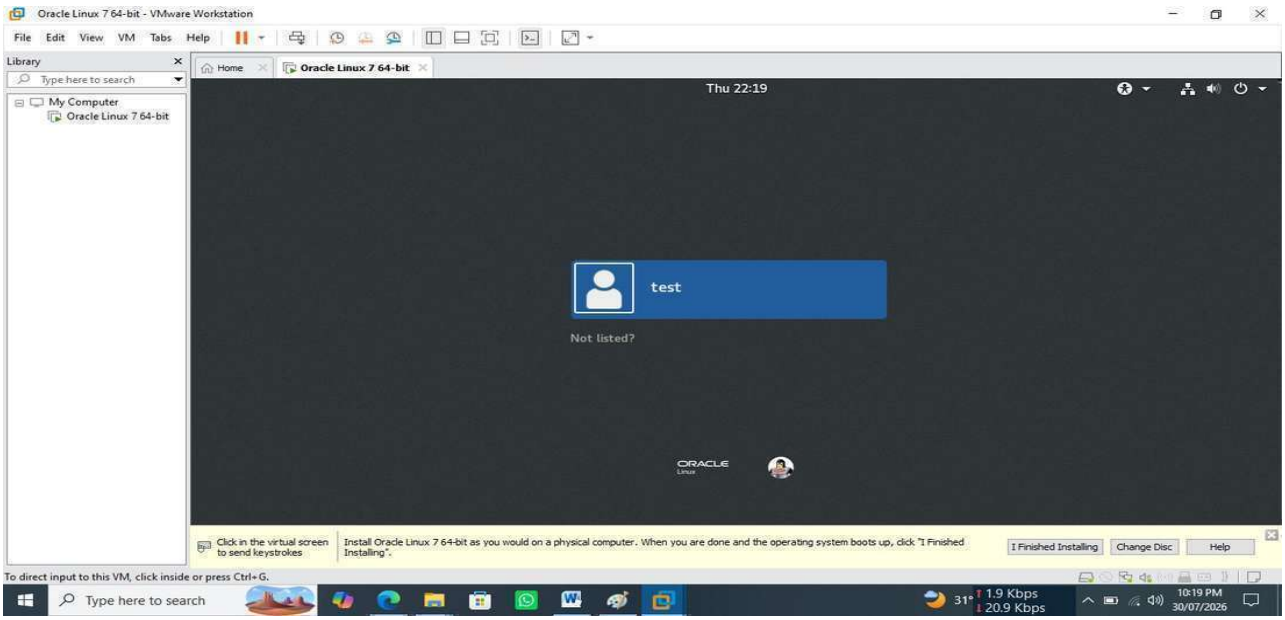



**Figure 2.10: Oracle Linux Login Screen**



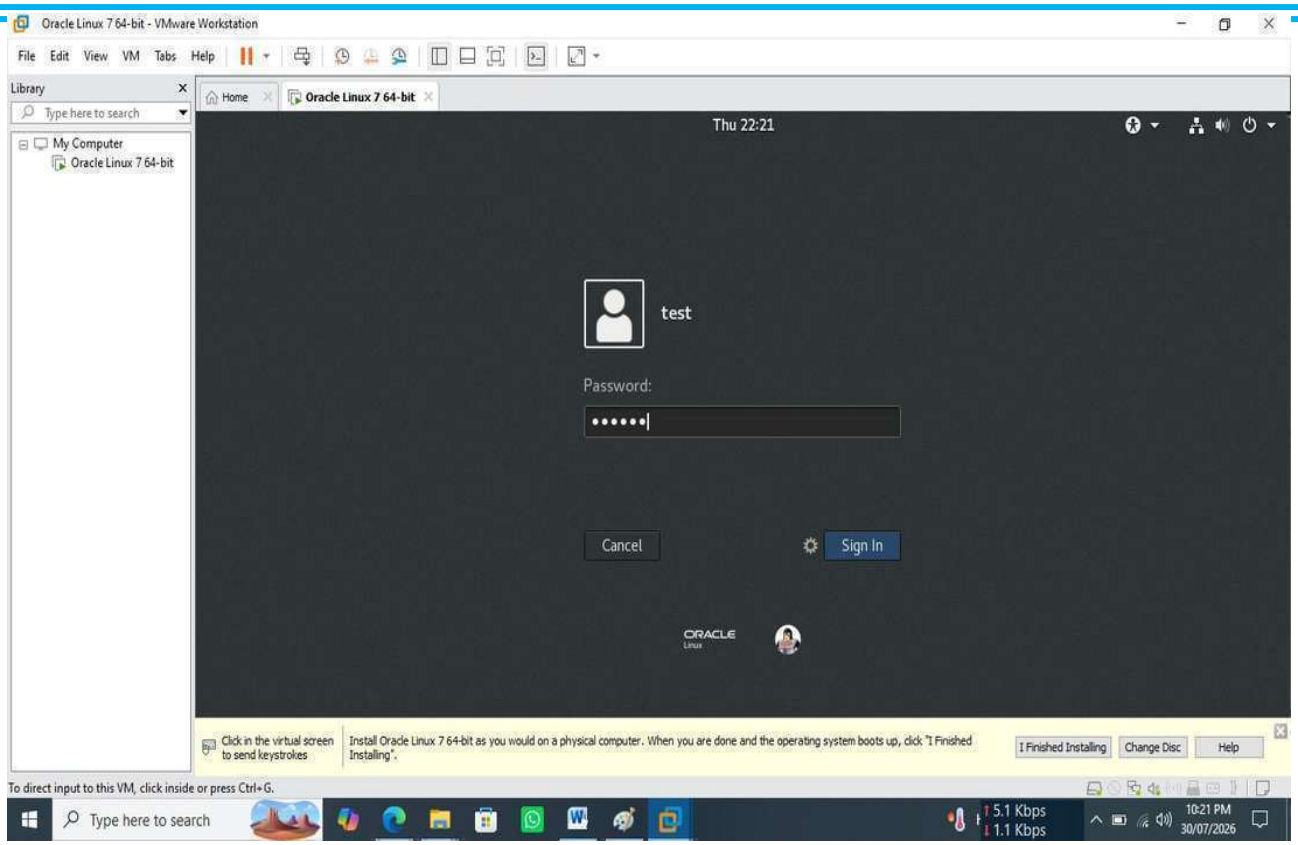
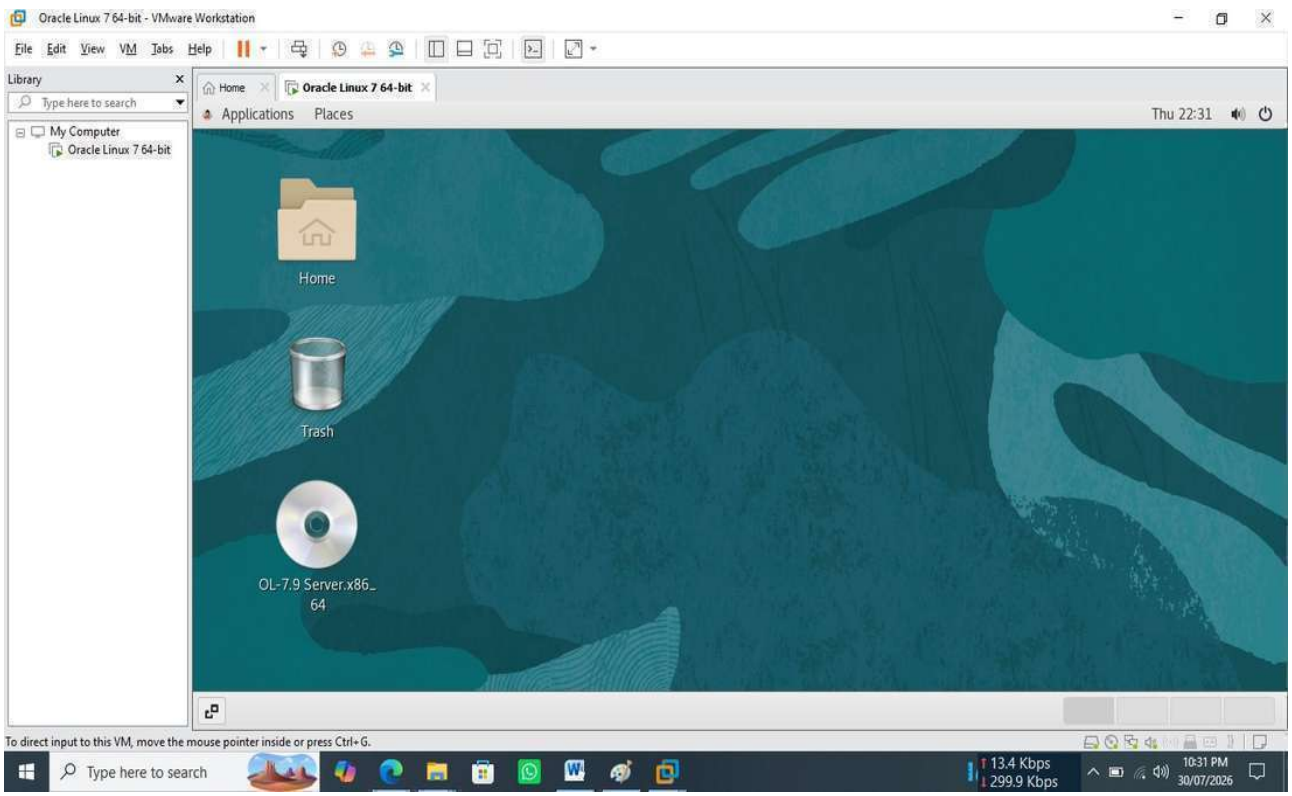


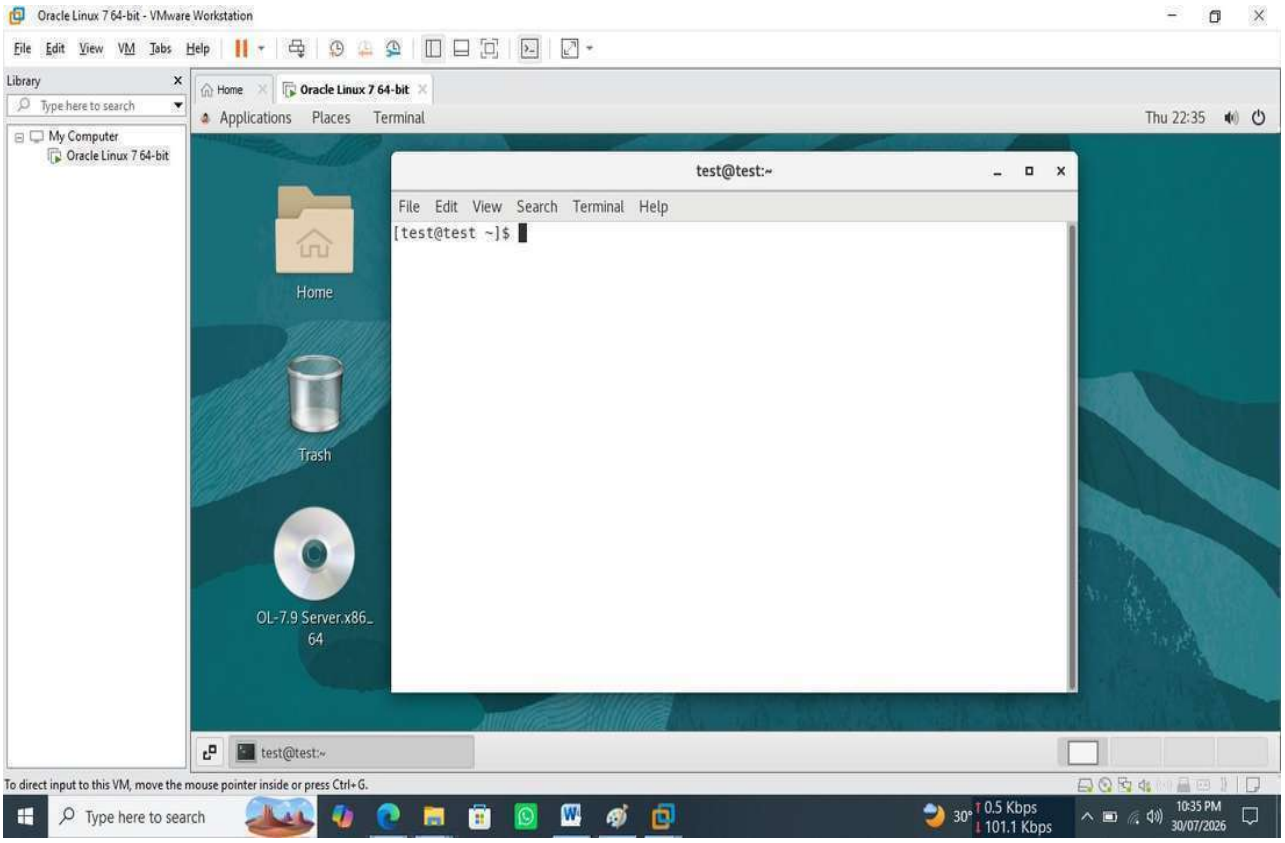

**Figure 2.11: Oracle Linux Desktop Environment**




### 2.2.8 Step 7: Verification

Finally, the Oracle Linux desktop environment was displayed without any errors. This confirmed that the operating system had been installed successfully and was ready for Oracle Database installation in the next phase of the internship.

**Figure 2.12: Oracle Linux Running Successfully**




***** Best of Luck *****